

SA04 MONITOR AIR AND WATER QUALITY | O (MAX: 1 PT)

Intent: Assess indoor air quality and water quality, which contribute to the health and well-being of building users.

Summary: This feature requires projects to monitor air quality and water quality levels at least once per year.

Issue: Exposure to air pollutants such as Volatile Organic Compounds (VOCs), ozone, particulate matter, carbon monoxide and others has been shown to increase the risk of respiratory and cardiovascular diseases in addition to causing thousands of cancer deaths annually.²²⁹ Inhaling pollutants present symptoms ranging from headaches, dry throat, eye irritation and runny nose that may later develop into extreme health outcomes such as asthma attacks and cancer.^{194,230,231} In addition, radon exposure is the second cause of lung cancer, after tobacco use.²³² Therefore, it is important to define indoor air quality levels that minimize risk to human health. All water systems require some degree of validation to ensure that health targets are met under their operational conditions, for which sampling is an effective verification tool.²²⁷

Solutions: The World Health Organization (WHO) and other regulatory bodies such as the U.S. Environmental Protection Agency (EPA) identify a list of “criteria” air pollutants and have established permissible levels for such criteria pollutants based on epidemiological studies that show the relationships between concentrations of these pollutants, duration of exposure and health risks. Achieving the goal of clean indoor air as defined by permissible levels requires the joined efforts of both professionals and building users in the implementation of adequate approaches. Indoor air quality can be properly managed through different features listed in the WELL Air concept, including source control strategies, passive and active building design and operation strategies and human behavior interventions.^{233,234} Basic management for water quality ensures that the water sourcing, treatment and delivery operates as designed. Turbidity, residual chlorine and pH monitoring help control basic chemical and microbiological water characteristics and, if trends are detected, may inform that actions may be needed to protect the quality of the water, such as changing a filter, checking the building’s pipes for leaks or inquiring with local relevant governmental authorities.

Part 1 Part 1 (Max: 1 Pt)

For All Spaces except Dwelling Units:

The following pollutants are monitored in occupiable spaces (with a quantity and location of sampling points complying with the requirements outlined in the Performance Verification Guidebook) at intervals no longer than once per year, and results are submitted annually through the WELL digital platform:

- a. Ozone.
- b. Carbon Monoxide.
- c. PM_{2.5} and/or PM₁₀.
- d. Total VOCs and/or Formaldehyde.

The following requirements are met:

- a. The following parameters are sampled at drinking water dispensers in occupiable spaces at intervals of no less than once per year:
 1. Turbidity.
 2. pH.
 3. Residual (free) chlorine.
 4. Total coliforms, only if residual chlorine is below detection limits.
- b. Tests are required at 5% of drinking water dispensers, up to a maximum of four tests.
- c. The water quality results are submitted annually through the WELL digital platform.